

Online Appendix for “Artificial Intelligence, Firm Growth, and Product Innovation”

A1 Model of AI and Firm Growth

We outline a simple theoretical framework with heterogeneous multiple-product firms and both process and product innovation. The framework extends the models of industry dynamics in [Bernard et al. \(2010\)](#) and [Melitz and Redding \(2014\)](#). We aim to introduce the simplest model necessary to capture the essence of a broad class of models featuring product and process innovation and derive the empirical predictions that we can test using our data. The key feature of the model is that investments in AI can help firms grow either through product innovation (increasing the appeal of their existing or future products) or through process innovation (improving the productivity of their existing products).

Consider a continuum of products $i \in [0, 1]$, with the representative consumer’s preferences taking the Constant Elasticity of Substitution (CES) form:

$$U = \left[\int_0^1 (a_i C_i)^{(v-1)/v} d_i \right]^{\nu/(\nu-1)}, \quad \nu > 1,$$

where C_i is consumption of product i , and a_i captures the relative importance of product i .

For each product i , there is a set of horizontally differentiated varieties, produced by different firms. Preferences across these varieties for any product also take the CES form:

$$C_i = \left[\int_{f \in F_i} (\lambda_{i,f} c_{i,f})^{(\sigma-1)\sigma} df \right]^{\sigma/(\sigma-1)}, \quad \sigma > 1,$$

where f indexes firm varieties within products, and F_i is the set of all firms that produce varieties of product i . The demand parameter $\lambda_{i,f}$ determines the relative demand for firm f variety of the product. For brevity, we assume that the elasticity of substitution across varieties of a product is the same for all products. We also assume that elasticity of substitution across varieties within products is greater than the elasticity of substitution across products (i.e., $\sigma > \nu$), since consumers are likely to first form demand for a specific product and then search for varieties within that product.

When a firm enters, it draws its initial productivity in each product, $\varphi_{i,f} \in [\underline{\varphi}, \bar{\varphi}]$, and consumer tastes for the characteristics of its product variety in each product, $\lambda_{i,f} \in [\underline{\lambda}, \bar{\lambda}]$. While $\varphi_{i,f}$ captures the firm’s production efficiency, $\lambda_{i,f}$ determines the product appeal (for example, a firm with a

higher-quality product has a higher $\lambda_{i,f}$). Both productivity $\varphi_{i,f}$ and consumer tastes $\lambda_{i,f}$ are firm-product-specific and differ across various products produced by the firm.⁴⁷

The production technology takes the following form. A firm incurs fixed corporate headquarters costs of M_h units of labor, which is irrespective of the number of products produced, and a fixed cost of M_i units of labor for each product i that it produces. Both fixed costs are homogeneous across firms. Labor has unit cost w . Within each product i , each firm supplies a distinct horizontally-differentiated variety. There is a constant marginal cost of producing a variety, which is inversely proportional to firm productivity $\varphi_{i,f}$. The total amount of labor required to produce $q_{i,f}$ units of a variety i is therefore:

$$l_{i,f} = M_i + \frac{q_{i,f}}{\varphi_{i,f}}.$$

Because of monopolistic competition, each firm chooses its price to maximize its profits subject to a downward-sloping residual demand curve with constant elasticity σ . The equilibrium price for each variety is a constant mark-up over marginal cost:

$$p_{i,f}(\varphi_{i,f}, \lambda_{i,f}) = \frac{\sigma}{\sigma - 1} \frac{w}{\varphi_{i,f}}.$$

Firm profits for product i are therefore:

$$\pi_{i,f}(\varphi_{i,f}, \lambda_{i,f}) = \frac{(\sigma - 1)^{\sigma-1}}{\sigma^\sigma} R_i \left(\frac{\varphi_{i,f} \lambda_{i,f} P_i}{w} \right)^{\sigma-1} - w M_i,$$

where $P_i = \left[\int_{f \in F_i} \left(\frac{p_{i,f}(\omega)}{\lambda_{i,f}(\omega)} \right)^{1-\sigma} df \right]^{1/(1-\sigma)}$ is the price index for product i , and R_i is an index of market demand that proportionally scales every firm's residual demand and does not vary across firms within a product. With a continuum of firms, each firm is of measure zero relative to the market as a whole, and correspondingly its production in any product has measure zero relative to all firms producing that product.

A firm observes its productivity and consumer tastes before deciding whether to produce each product. Firm f produces product i if and only if $\pi_{i,f}(\varphi_{i,f}, \lambda_{i,f}) > 0$. Profits are increasing in productivity $\varphi_{i,f}$ and product appeal $\lambda_{i,f}$, therefore firms with higher productivity and product appeal for a product are more likely to actively produce the product. Suppose Φ_f is the set of products with positive profits for a firm, then the firm's total profit is:

$$\pi_f = \int_{i \in \Phi_f} \pi_{i,f}(\varphi_{i,f}, \lambda_{i,f}) di - w M_h.$$

⁴⁷Bernard et al. (2010) assume that productivity is common across products within firms, but we relax this assumption.

Process Innovation and Productivity Improvements. When a firm invests in artificial intelligence technologies, it may lead to process innovation and higher production efficiency. We introduce process innovation in the model similarly to [Bustos \(2011\)](#). A firm with productivity $\varphi_{i,f}$ can pay additional fixed costs to increase the productivity of all products that it is actively producing to $\iota\varphi_{i,f}$, where $\iota > 1$ is the proportional productivity increase. The technology adoption choice involves a tradeoff between a fixed cost and a per-unit profit increase. For each product the firm produces, costs decrease proportionately, which increases the firm's profits from that product. Importantly, process innovation does not change the productivity of products that the firm does not actively produce and therefore does not change the set of products Φ_f .

In this model, process innovation also increases revenue per worker and profit per worker. In particular, the revenue per worker for product i is:

$$\frac{p_{i,f}q_{i,f}}{l_{i,f}} = \frac{\sigma}{\sigma - 1} \frac{w}{1 + M_i\varphi_{i,f}/q_{i,f}},$$

which is increasing in $\varphi_{i,f}$ since $\varphi_{i,f}/q_{i,f} \propto \varphi_{i,f}^{1-\sigma}$ is decreasing in $\varphi_{i,f}$. The profit per worker is:

$$\frac{\pi_{i,f}}{l_{i,f}} = \frac{\frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} R_i \left(\frac{\lambda_{i,f} P_i}{w} \right)^{\sigma-1}}{R_i \left(\frac{\sigma-1}{\sigma w} \right)^\sigma (\lambda_{i,f} P_i)^{\sigma-1} + M_i \varphi_{i,f}^{1-\sigma}} - w \frac{M_i}{l_{i,f}},$$

which is increasing in $\varphi_{i,f}$ since $\sigma > 1$ and $l_{i,f}$ is increasing in $\varphi_{i,f}$.⁴⁸

Product Innovation. Investments in artificial intelligence technologies may also affect product innovation. Suppose that a firm can conduct product innovation to produce products with higher than existing consumer tastes (or product appeal). More specifically, by investing $\psi_f x^2$ units of labor, $x \in [0, 1]$ fraction of all the products are randomly drawn, which may include products that the firm already produces and products that the firm is not actively producing. For these randomly drawn products, the firm's existing consumer tastes $\lambda_{i,f}$ are multiplied by $\kappa > 1$. Product innovation can increase firms' revenue in two ways: first, it may increase the product appeal and revenue of existing products; second, by increasing the product appeal of products which are unprofitable for the firms to produce originally, it may also expand Φ_f , the set of products that the firm produces with positive profits. The change in productivity is ambiguous: since the productivity $\varphi_{i,f}$ is product-specific, expanding the set of products may increase or decrease the overall productivity.

Artificial intelligence can potentially reduce the cost of product innovation ψ_f and spur firm growth, as discussed in Section 1. First, the ability of AI to analyze large datasets and make pre-

⁴⁸In more general models with multiple inputs and decreasing returns to scale, process innovation could also lead to a decrease in revenue per worker.

dictions can reduce the cost of experimentation, which is key to product development. Second, firms can improve the quality of products and services by building AI models directly into products. Third, AI can reduce the costs of product customization by learning more efficiently about customers' tastes and needs from big data, which help improve product appeal to customers.

The two channels—productivity improvements and product innovation—both lead to higher firm growth in revenue but have very different predictions. Process innovation leads to lower costs and higher productivity, while the effects of product innovation on the firm's total costs and productivity are ambiguous. On the other hand, product innovation leads to the creation of new products and improvements of existing products, while process innovation does not affect the scope of products firms produce.

A2 Case Studies on Firms' AI Investments

In order to illustrate the wide range of applications of AI technologies by individual firms, we provide detailed summaries of the investment patterns and uses of AI technologies within four firms in four different industries.

A2.1 UnitedHealth Group

UnitedHealth Group (UNH) is a large managed healthcare company based in Minnetonka, Minnesota. The group includes a healthcare arm (UnitedHealthcare) established in 1977 and a new technology arm founded in 2011 (Optum). While the UnitedHealthcare arm makes use of AI techniques to optimize operations ranging from cost projections to fraud detection in medical claims, the launch of Optum highlights the way in which firms such as UNH can leverage AI technologies to expand operations by creating new products and entering new market segments. UNH is one of very few companies with access to detailed patient, patient-physician, and drug-patient interaction data for large portions of the U.S. and many additional global locations, making it perfectly placed to harness AI in its operations.

AI use cases and product impact. Most of the AI investments and impact at UNH center around its Optum arm. The traditional UnitedHealthcare part of UNH uses AI in a limited capacity for predictive analytics that inform business decisions and safeguards for vulnerabilities such as fraud detection. The launch of Optum in 2011 has enabled UNH to leverage AI technologies to deliver new products across several healthcare markets. At its core, Optum is a vast data store of proprietary and 3rd party datasets linked together to enable machine-learning-based analysis. Specifically, the AI-powered Optum products include: (i) statistics on drugs and potential alternatives through the pharmaceutical platform Optum Rx; (ii) analysis of electronic medical records

through the Optum One platform for physicians; and (iii) the Optum Population Health Management platform for larger institutions (including employers and federal and state agencies) to optimize costs and accessibility to care. The AI-powered OptumIQ system, which is leveraged throughout the Optum solutions, also targets machine-learning-based prediction and diagnostics for diseases such as atrial fibrillation.

Timeline of AI investments at UNH. The use of AI technologies at UNH traces further back than at most firms. As early as the 1990s, UNH piloted AdjudiPro, an AI-powered platform for processing claims from physicians. However, the presence of AI-skilled labor at UNH remained low throughout the 1990s and 2000s, noticeably picking up in 2011 with the launch of the Optum platform. Thereafter, UNH's investment in AI human capital rose steadily throughout the 2010s. The Optum arm of the firm released the Optum360 and Impact Pro products in 2013 and the Optum One Analytics Platform in 2014, prompting a further acceleration in the rise of UNH's AI human capital in the second half of the decade. The timeline of AI investments at UNH is displayed in Figure A1.

Internal structure of UNH's AI workforce. UNH has a centralized approach to AI integration, with strategic decisions primarily coming from the headquarters in Minnesota and regional offices handling specific applications. Correspondingly, the majority of UNH's AI workforce concentrates in Minneapolis and Minnetonka, including senior personnel heading AI and machine learning efforts, automation/deployment, consumer analytics, and Optum enterprise analytics. Locations outside of the headquarters tend to employ predominantly engineering and general IT personnel to support the AI efforts.

A2.2 JPMorgan Chase & Co

JPMorgan Chase & Co (JPM) is the largest bank in the U.S., based in New York City, NY, with consumer banking that has relationships with more than half of U.S. households, a commercial banking arm, a large investment banking business, and a sizable asset management arm. The bank stores hundreds of petabytes of data ranging from credit card transactions and loan applications, to financial news and market data, to alternative data sources.

AI use cases and product impact. The main use cases for AI at JPM fall into the following categories: (i) risk modeling and management ranging from internal cybersecurity to fraud detection in consumer banking and assessment of geo-political risks; (ii) quantitative analysis and algorithmic investment products, including the Algo Central, LOXM, and DeepX programs aimed at executing trades at both maximal speed and optimal prices; (iii) general analytics for Big Data use in broad internal applications including recruiting; and (iv) product development, including enhancements to mobile apps and customer support through AI-powered virtual assistants. In addition, JPM also employs AI in more peripheral applications, for example, with methods for

processing of alternative data such as satellite images and mapping contingency plans for AI-driven workforce disruptions. The use of AI at JPM is aimed at both cutting costs (e.g., through risk assessment) and creating new products (e.g., machine-learning-powered trading platforms such as DeepX).

Timeline of AI investments at JPM. As highlighted by Figure A2, investments in AI at JPM began at the turn of the century, with a steady increase through the first decade turning into an exponential growth in the second decade. The explosion in AI investments at JPM during the 2010s is marked by the acquisition of the multimedia recommendations patent in 2011; an underscoring of the risks associated with data security following a data leak in 2016; and finally the establishment of a dedicated AI research initiative (Machine Learning Center for Excellence) spearheaded by Dr. Manuela Veloso (previously the Chair of the Machine Learning Department at Carnegie Mellon University) in 2018.

Internal structure of JP Morgan's AI workforce. AI efforts at JPM are centered in the New York location, with peripheral AI expertise throughout the U.S., in London, and in India. JPM has taken a top-down approach to AI investments, with involvement from the highest levels of management and the establishment of a dedicated AI research team in 2018. At the same time, JPM's investments in AI have seen not only the formation of dedicated AI hubs, but also a different approach to corporate structuring. Specifically, the firm's approach relies heavily on small skilled and responsive AI "task-forces" specializing in different sectors (quantitative analysis, user experience, etc.), which can alternatively work on experimental projects (e.g., satellite imagery analysis) or coordinate together to work on core applications (trading algorithms, firm-wide cybersecurity).

A2.3 Caterpillar Inc.

Caterpillar Inc. is a large construction manufacturing firm headquartered in Deerfield, IL, with a variety of additional business activities including financial products and insurance. The firm has correspondingly varied applications for AI, ranging from inventory management to part recognition, to credit scoring for machinery financing.

AI use cases and product impact. AI investments at Caterpillar are organized along several key verticals. First, the Data Innovation Lab at UIUC conducts core projects in demand forecasting (unstable demand anticipation) and inventory management, in part identification (using techniques from image recognition), and in tracking and tracing technology for fleet management. Second, Caterpillar's asset intelligence efforts include a product line of Internet of Things (IoT) style analytics for managers and machine operators, which facilitates data collection, interpretation, predictive maintenance, and integration. Lastly, smaller targeted efforts at Caterpillar also employ AI techniques in other parts of the business, including leveraging sensor-based data for equipment management and using drone data to optimize job site organization. Caterpillar's uses

of AI serve to modernize the firm's machinery, streamline operations and reduce waste through better forecasting and inventory management, and expand the product offerings with the IoT product line and efficient long-term service contracts.

Timeline of AI investments at Caterpillar. Caterpillar began employing workers with AI expertise at the turn of the century, but the growth in the firm's AI workforce went hand-in-hand with the growth in the firm's overall workforce throughout the 2000s (with a dip during the financial crisis). The share of AI employees at Caterpillar noticeably picked up only in mid-2010s, with the CEO Douglas Oberhelmer underscoring the importance of capitalizing on the firm's vast available data resources. Since 2014, Caterpillar has aggressively pursued the development of "smart" machinery, connecting it to predictive IoT-style networks and developing better models for demand prediction. In 2015, Caterpillar established the Analytics and Innovation Division headed by Greg Foley, and in 2016, the firm hired Morgan Vawters as the Chief of Analytics. The timeline of Caterpillar's investments in AI human capital is presented in Figure A3.

Internal structure of Caterpillar's AI workforce. The majority of the AI employees at Caterpillar are in the firm's Technology division, with notable presence also in Business and Production departments. The major locations setting the trend for Caterpillar's AI adoption are the company centers in Chicago and Peoria, Illinois, with projects percolating through the dedicated research centers such as the Champaign Innovation Center and production centers such as the manufacturing plant in Aurora, Illinois.

A2.4 Qualcomm Inc.

Qualcomm Inc. is a wireless telecommunications firm headquartered in San Diego, CA. The firm produces a number of products including semiconductors, hardware, software, and other services related to wireless technology. Device manufacturers such as Apple are Qualcomm's primary clients.

AI use cases and product impact. The principal use of AI at Qualcomm over the past decade and a half has been the improvement of its core products. This includes optimization of chips within mobile devices, improvements to the camera using techniques from computer vision for face recognition and auto-adjustments, audio and video processing, physical sensitivity, power use, and location tracking capabilities. More recently, Qualcomm made a large investment in the development of the Snapdragon Neural Processing Engine (SNPE) platform, which offers a combination of hardware and software on android devices that allows developers to more easily create AI-powered or assisted applications. With the exception of a few stand-alone projects for internal data processing efficiency (e.g., improving internal servers), Qualcomm does not appear to be heavily invested in applying AI for applications such as sales or supply chain optimization, unlike Caterpillar Inc. described above.

Outside of its core businesses, Qualcomm has invested in a number of side products at more exploratory or proof-of-concept stages, such as general work on autonomous vehicles, or enterprise partnerships, for example with Accenture and Kellogg on virtual reality tracking of customers for marketing purposes. This highlights the broad scope of AI technologies that facilitate firms entering new markets: for example, the autonomous vehicle work at Qualcomm makes use of the efforts aimed at enhancing smartphone components, only applied to a different domain.

Timeline of AI investments at Qualcomm. As can be seen from the timeline in Figure A4, the presence of AI employees at Qualcomm began earlier than in the other firms, and by 2007 the firm initiated dedicated AI research projects in its research arm. The ramp up continued through 2013, marked by collaborations with outside partners such as Brain Corp and internal projects on problems such as face detection. After 2013, Qualcomm saw notable consequences of the earlier investments, including the first release of SNPE and the formation of an organizationally separate AI research group, but the share of Qualcomm's overall workforce that is skilled in AI remained approximately flat from 2013 to 2018.

Internal structure of Qualcomm's AI workforce. Between 2000 and 2018, the majority of Qualcomm's AI employees have been engineers focused on the improvement of the core product being developed at each point in time, supported by an auxiliary staff of patent counsels and data scientists. In 2018, Qualcomm established a separate AI research group, which is bringing about increased centralization of its AI workforce. Specifically, AI efforts at Qualcomm are organized around the San Diego headquarters, with leadership on overall AI strategy, the newly formed AI research group, and teams spanning nearly every project from computer vision R&D to GPU architecture. Smaller AI offices, scattered mostly throughout the U.S. and Canada, tend to focus on single elements of Qualcomm's AI initiative (for example, SNPE in Toronto and positioning sensors in Santa Clara).

Figure A1. Timeline of AI investments by UnitedHealth Group

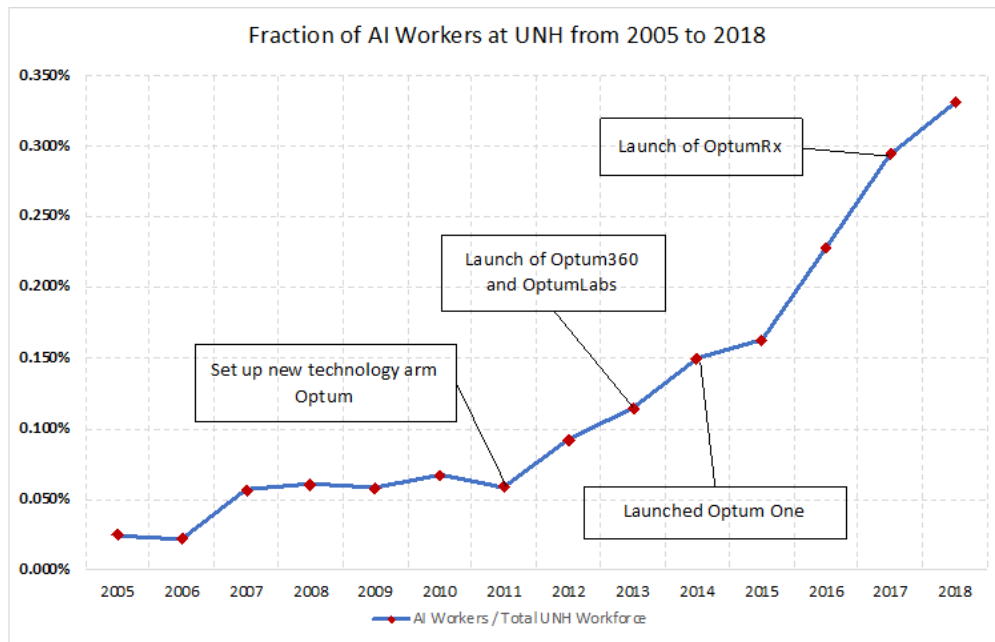


Figure A2. Timeline of AI investments by JPMorgan Chase & Co

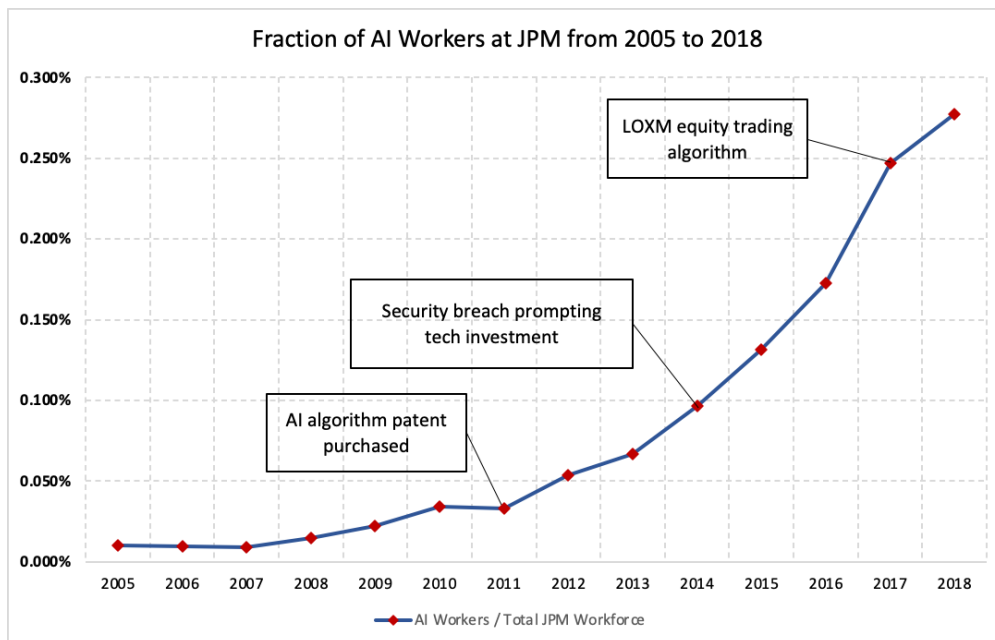


Figure A3. Timeline of AI investments by Caterpillar Inc

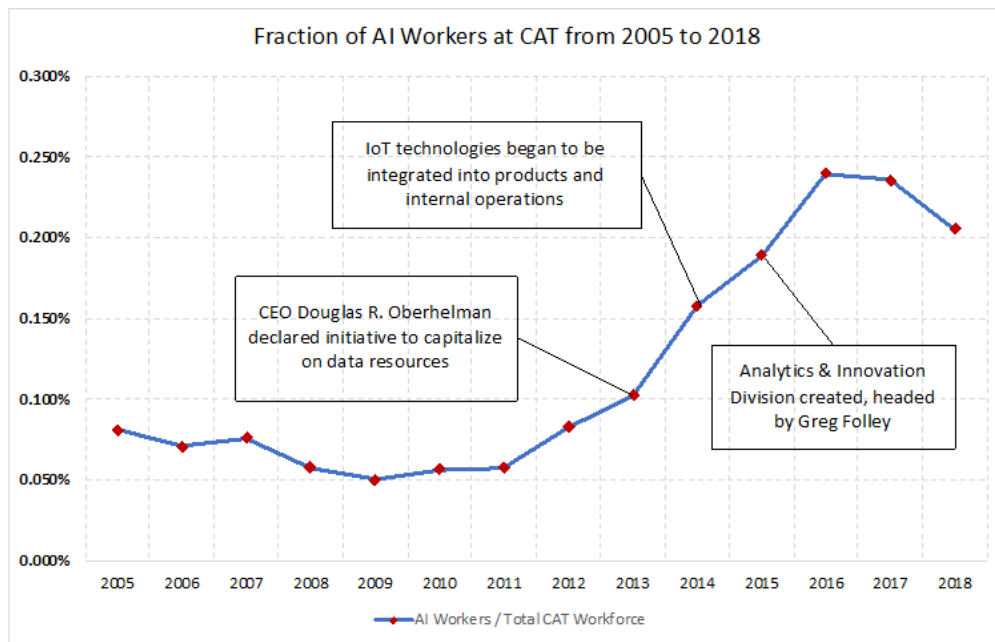
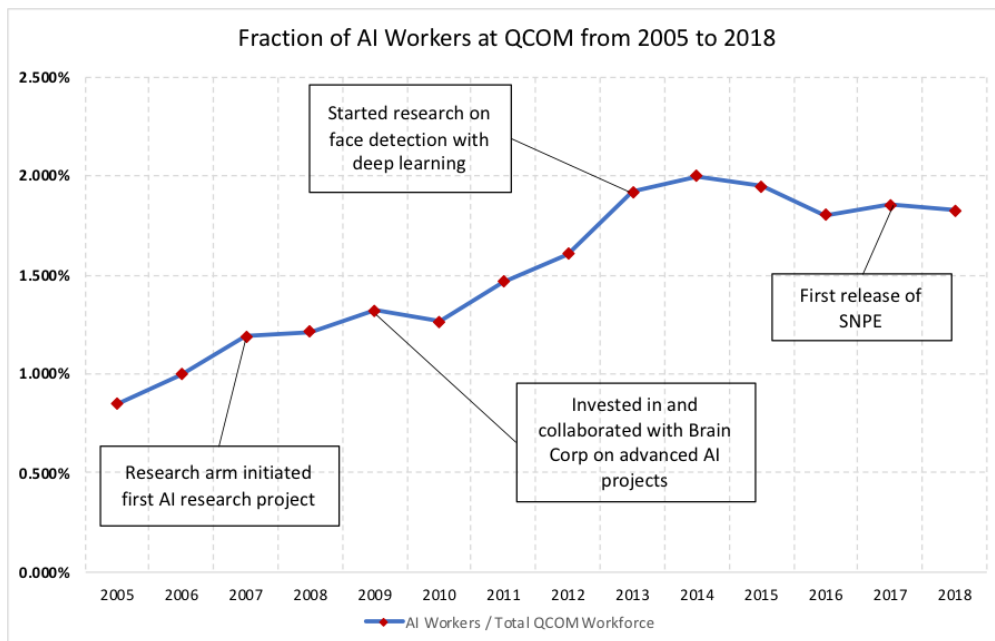


Figure A4. Timeline of AI investments by Qualcomm Inc



A3 Additional Figures and Tables

Figure A5. Matching Rate to Compustat in Job Postings Data

This figure shows the time series of the share of all job postings and the share of AI job postings (job postings with continuous measure ω_j^{AI} above 0.1) that are matched to Compustat firms in the Burning Glass data in 2007 and in each year from 2010 to 2018.

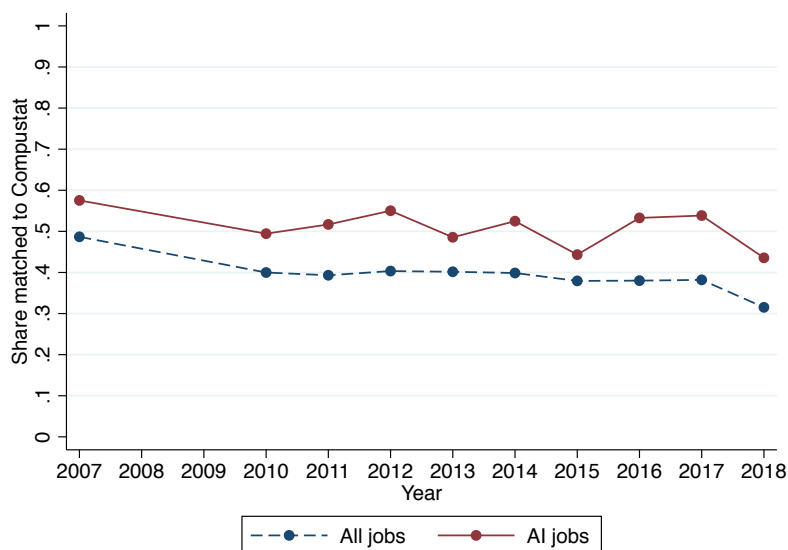


Figure A6. Distribution of AI Investments across U.S. Geographies

This figure shows a heat map of changes in the job-posting-based measure of AI investments across geographies in the U.S. The figure plots the change in the average AI-relatedness measure (w_j^{AI}) of job postings of public firms in each commuting zone from 2010 to 2018.

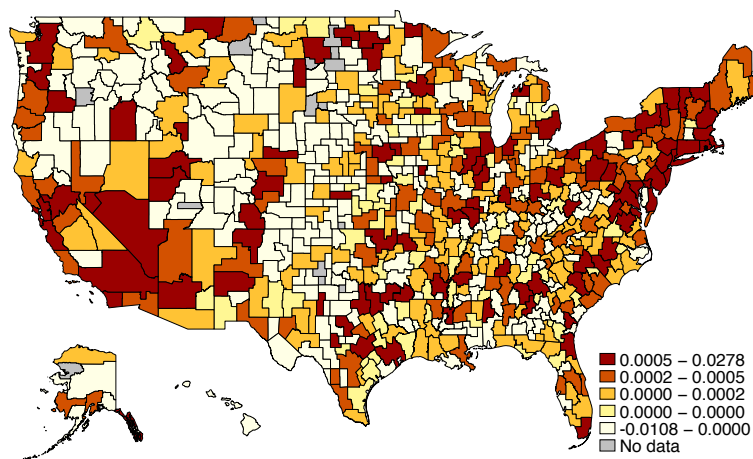
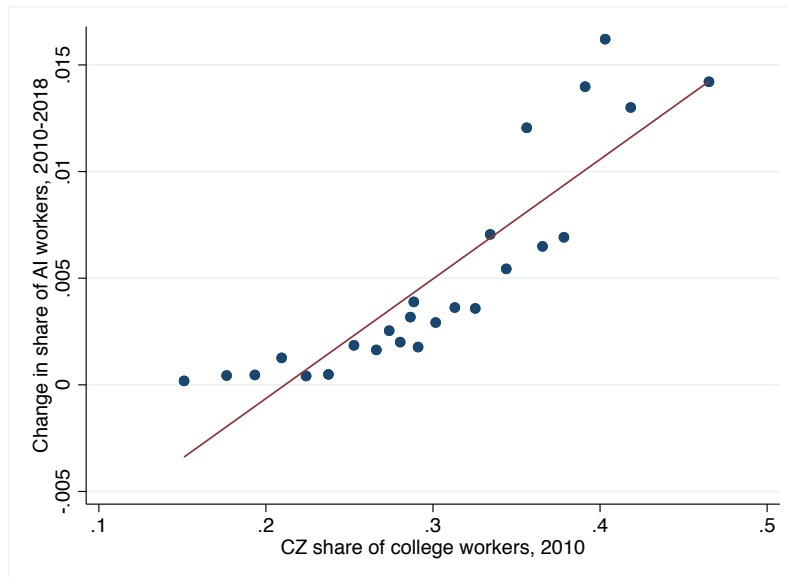


Figure A7. AI Investments and Local Conditions

This figure presents binned scatterplots of commuting-zone-level AI investments against local conditions. Solid lines are the fitted regression lines, where the regressions are weighted by commuting zones' populations in 2010. The y-axis is the change in AI investments (i.e., the change in the share of AI workers) from 2010 to 2018, using the Burning Glass data (based on the sample of public firms). The x-axis in Panel (a) is the average log wage of a commuting zone in 2010. The x-axis in Panel (b) is the share of college educated workers in a commuting zone in 2010. The log wage and the share of college-educated workers are from the Census American Community Survey. The t-statistic on the regression slope is 23.8 in Panel (a) and 24.1 in Panel (b).



(a) AI Investments and Local Average Wage



(b) AI Investments and Local Share of College-educated Workers

Figure A8. Relationship between the Instrument and Firm Growth

This figure presents binned scatterplots of firm growth from 2010 to 2018 against the instrument – ex-ante firm-level exposure to AI-skilled graduates from AI-strong universities (see the definition of the instrument in Section 4.3 and the details of instrument construction in Appendix A). Solid lines are the fitted regression lines. For firm growth, we consider changes in log sales in Panel (a), log employment in Panel (b), and log market value in Panel (c). The observations are weighted by the number of Cognism resumes in 2010. All specifications control for industry sector fixed effects, ex-ante exposure to universities that are strong in computer science research as well as top 10 universities, baseline controls measured as of 2010 (firm-level log sales, cash/assets, firm age, and log number of resumes; log industry wage; characteristics of the commuting zones where the firms are located), as well as firm-level changes in log sales and log employment from 2000 to 2008.

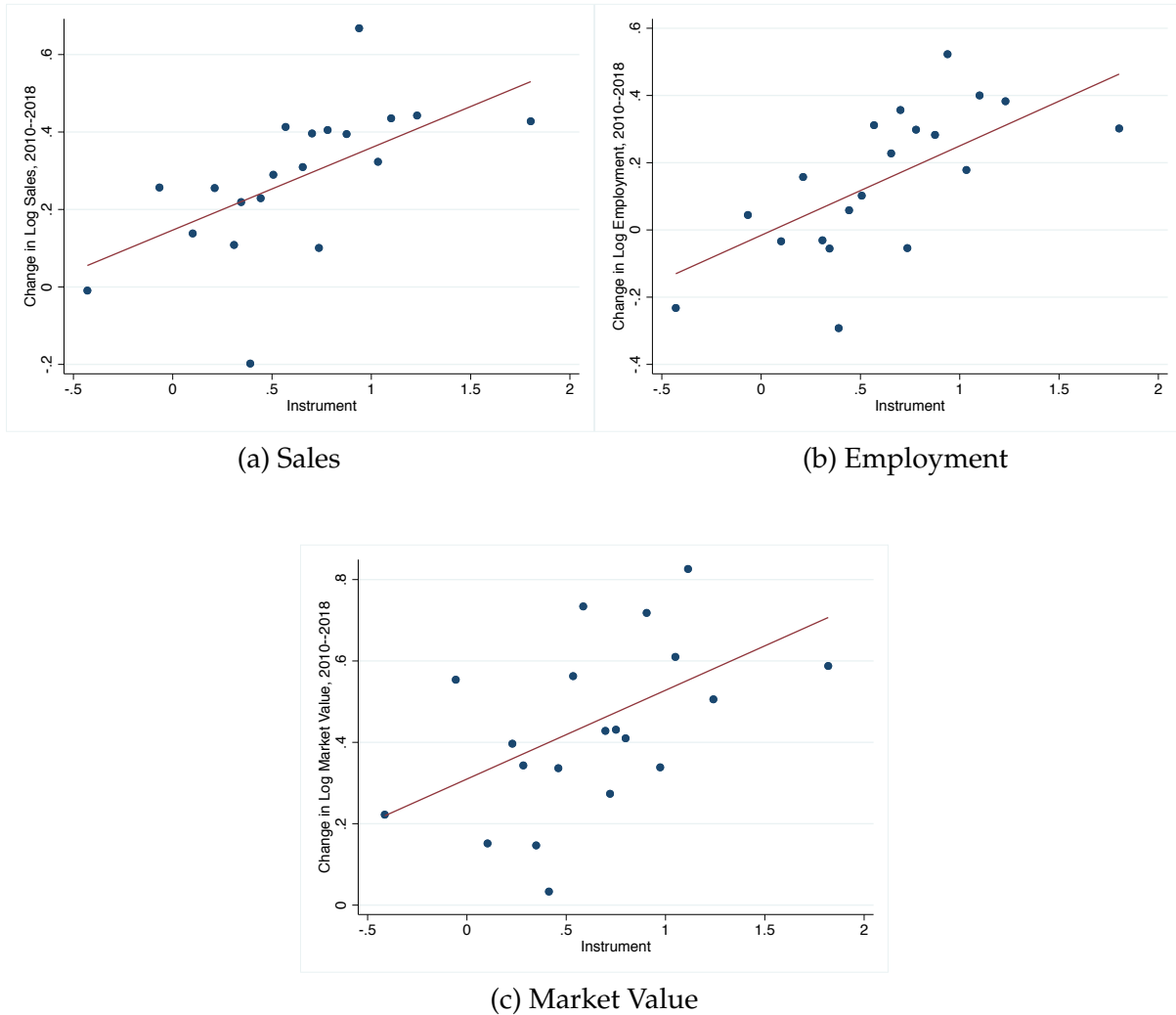


Table A1. Skills with Highest AI-Relatedness Measures in Burning Glass Job Postings

This table lists the top skills in the Burning Glass data ranked by the skill-level AI measure w_s^{AI} . For each skill, we report the percentage of jobs requiring that skill that also require one of the four core AI skills—artificial intelligence, machine learning, computer vision, and natural language processing. For example, for jobs that require “Recurrent Neural Network (RNN),” 96.5% also require one of the four core AI-skills. Only skills that appear in at least 50 job postings are included.

#	Skills	AI-relatedness Score
1	Artificial Intelligence	1.000
2	Computer Vision	1.000
3	Machine Learning	1.000
4	Natural Language Processing	1.000
5	ND4J (software)	0.980
6	Kernel Methods	0.979
7	Microsoft Cognitive Toolkit	0.975
8	Xgboost	0.972
9	Sentiment Classification	0.971
10	Long Short-Term Memory (LSTM)	0.971
11	Libsvm	0.968
12	Semi-Supervised Learning	0.968
13	Recurrent Neural Network (RNN)	0.965
14	Word2Vec	0.956
15	MXNet	0.953
16	Caffe Deep Learning Framework	0.950
17	Autoencoders	0.949
18	MLPACK (C++ library)	0.942
19	Keras	0.941
20	Theano	0.938
21	Torch (Machine Learning)	0.932
22	Wabbit	0.929
23	Boosting (Machine Learning)	0.905
24	TensorFlow	0.904
25	Vowpal	0.903
26	Convolutional Neural Network (CNN)	0.897
27	Jung Framework	0.894
28	OpenNLP	0.894
29	Natural Language Toolkit (NLTK)	0.892
30	Unsupervised Learning	0.891
31	Dlib	0.891
32	Scikit-learn	0.889
33	Latent Semantic Analysis	0.889
34	Latent Dirichlet Allocation	0.889
35	Stochastic Gradient Descent (SGD)	0.881
36	Gradient boosting	0.872
37	Dimensionality Reduction	0.861
38	Deep Learning	0.859
39	DBSCAN (Density-Based Spatial Clustering of Applications with Noise)	0.855
40	AI ChatBot	0.844
41	Recommender Systems	0.842
42	Random Forests	0.840
43	Deeplearning4j	0.839
44	Support Vector Machines (SVM)	0.817
45	Unstructured Information Management Architecture	0.806
46	Apache UIMA	0.805
47	Maximum Entropy Classifier	0.799
48	Hidden Markov Model (HMM)	0.796
49	Pybrain	0.786
50	Computational Linguistics	0.780
51	Naive Bayes	0.768
52	H2O (software)	0.763
53	Expectation-Maximization (EM) Algorithm	0.763
54	WEKA	0.761
55	Clustering Algorithms	0.740
56	Matrix Factorization	0.739
57	Object Recognition	0.727
58	Classification Algorithms	0.721
59	Information Extraction	0.709
60	Image Recognition	0.706
61	Bayesian Networks	0.705
62	Supervised Learning (Machine Learning)	0.695
63	OpenCV	0.688
64	K-Means	0.683
65	Sentiment Analysis / Opinion Mining	0.679
66	Machine Translation (MT)	0.655
67	Neural Networks	0.640

Table A2. Examples of AI and Non-AI Job Postings in Burning Glass

This table displays examples of job postings and their continuous AI measure ω_j^{AI} . Jobs 1–10 are examples of AI-skilled jobs, with the first five being non-data-specific and the last 5 being data-specific. Jobs 11–20 are examples of non-AI-skilled jobs, with the first five being data specific and the last 5 being non-data-specific. The AI relatedness score of each skill is listed in parentheses.

	Job Title	Employer	Skills	Score
	AI jobs			
1	Research Engineer - Natural Language Processing	InterActiveCorp	Machine Learning (1), Natural Language Processing (1), Natural Language Toolkit (0.895), Computational Linguistics (0.777), WEKA (0.760), Information Extraction (0.709), Mahout (0.593), Information Retrieval (0.360), Apache Hadoop (0.204), Lucene (0.188), SOLR (0.142), C++ (0.067), Software Engineering (0.043), Python (0.116), Lexical Semantics (0.625), Ontologies (0.326), Java (0.040), PERL Scripting Language (0.034), Relational Databases (0.024), SQL (0.023), Search Analytics (0.022), Shell Scripting (0.020), Web Analytics (0.012), Research (0.011), Online Research (0.010), Extensible Markup Language (0.010)	0.31
2	Computer Vision & Image Processing Researcher	Rambus Incorporated	Computer Vision (1), Object Recognition (0.725), OpenCV (0.689), Pattern Recognition (0.442), CUDA (0.362), Image Processing (0.179), Troubleshooting Technical Issues (0.006), C++ (0.067), Communication Skills (0.003), MATLAB (0.113), Self-Motivation (0.002), Optical System Design and Analysis (0.019), Research (0.011), Writing (0.004), OpenGL (0.117), Prototyping (0.042), Very Large Scale Integration (0.037), Creativity (0.007)	0.21
3	Algorithm Developer	IBM	Natural Language Processing (1), Machine Learning (1), IBM Watson (0.125), Java (0.040), Software Development (0.027), Candidate Generation (0.013), Creativity (0.007), Troubleshooting (0.003), English (0.002)	0.25
4	Senior Autonomous Vehicle Localization Software Engineer	Nvidia Corporation	Computer Vision (1), Deep Learning (0.859), Linear Algebra (0.187), OpenGL (0.117), C++ (0.067), Software Engineering (0.043), Geometry (0.009), Motor Vehicle Operation (0.006), Teamwork / Collaboration (0.005), Calibration (0.004)	0.230
5	Speech Recognition Scientist	Vocera Communications	Computational Linguistics (0.780), Automatic Speech Recognition (0.457), Speech Recognition (0.215), Experiments (0.045), Performance tuning (0.011), Research (0.011), Written Communication (0.003)	0.217
6	Data Scientist	Zappos	Machine Learning (1), Natural Language Processing (1), Boosting (Machine Learning) (0.902), Support Vector Machines (0.816), Naive Bayes (0.759), Matrix Factorization (0.738), Classification Algorithms (0.718), Data Science (0.379), Data Mining (0.159), NoSQL (0.119), Clustering (0.103), Data Structures (0.069), Relational DataBase Management System (0.028), SQL (0.023), Attribution Modeling (0.072), Detail-Oriented (0.002), Revenue Projections (0.003), Traffic Maintenance (0.002)	0.384
7	Data Mining Engineer	Apple Inc.	Artificial Intelligence (1), Natural Language Processing (1), Machine Learning (1), Unsupervised Learning (0.891), Supervised Learning (0.696), Mahout (0.593), Pattern Recognition (0.442), Apache Hadoop (0.204), Image Processing (0.179), Data Mining (0.159), NoSQL (0.119), Data Collection (0.008), Communication Skills (0.003), Java (0.040), Detail-Oriented (0.002), MATLAB (0.113), SQL (0.023), Network Engineering (0.007), Research (0.011), Python (0.116), Meeting Deadlines (0.002), R (0.248), Predictive Models (0.243)	0.309
8	Big Data Engineer	Socialwire	Machine Learning (1), Recommender Systems (0.843), MapReduce (0.285), Apache Hadoop (0.204), Big Data (0.196), Facebook (0.006), R (0.248), Pinterest (0.003), Writing (0.004), MATLAB (0.113)	0.290
9	Big Data Senior Data Scientist	AT&T	Machine Learning (1), WEKA (0.760), Clustering Algorithms (0.738), Mahout (0.593), Data Science (0.379), Big Data (0.196), Data Mining (0.159), Clustering (0.103), Simulation (0.028), Experimental Testing (0.039), R (0.248), SPSS (0.067), Creativity (0.007), SAS (0.053), Information Systems (0.007), Experiments (0.045), Presentation Skills (0.006), Research (0.011), Data Quality (0.025)	0.235
10	Data Scientist	Warby Parker	Natural Language Processing (1), Natural Language Toolkit (0.895), Random Forests (0.839), Pandas (0.498), Data Science (0.379), PIG (0.290), Apache Hadoop (0.204), Data Mining (0.159), Data Visualization (0.136), Tableau (0.074), Pentaho (0.058), NumPy (0.552), SQL (0.023), Python (0.116), Java (0.040), DevOps (0.039), Agile Development (0.030), Creativity (0.007), Django (0.039), Apache Webserver (0.034), Predictive Models (0.243), Relational Databases (0.024), Data Modeling (0.037)	0.249
	Non-AI jobs			
11	Director Of Business Intelligence	Odesus Incorporated	Data Science (0.379), Data Transformation (0.060), SQL (0.023), Communication Skills (0.003), SQL Server Reporting Services (0.009), SQL Server (0.009), SQL Server Analysis Services (0.034), Budgeting (0.001), Microsoft Sharepoint (0.002), Data Warehousing (0.025), MySQL (0.028), Key Performance Indicators (0.006), Problem Solving (0.005), Web Analytics (0.012), Market Research (0.006), Data Modeling (0.037), Business Intelligence (0.026), Creativity (0.007)	0.037
12	Director, Data & Analytics	Decision Resources	Big Data (0.196), Business Intelligence (0.026), Business Intelligence Industry Knowledge (0.020), Teamwork / Collaboration (0.005), Biopharmaceutical Industry Knowledge (0.004), Communication Skills (0.003)	0.042
13	Senior Healthcare Economics Data Analyst	UnitedHealth Group	Tableau (0.076), Advanced Statistics (0.149), SAS (0.053), Data Analysis (0.026), SQL (0.023), Economics (0.016), Database Design (0.014), Clinical Data Analysis (0.012), Clinical Data Review (0.010), Business Process (0.006)	0.039
14	Data Analyst	United Technologies Corporation	Data Analysis (0.026), Data Quality (0.025), Data Management (0.018), Database Design (0.014), Proposal Writing (0.007), Product Improvement (0.007), Business Planning (0.002)	0.014
15	Sas Database Administrator	Pitney Bowes	SAS (0.053), SQL (0.023), Business Strategy (0.009), Teradata DBA (0.005), Self-Starter (0.004), Database Administration (0.004), Pivot Tables (0.004), Market Analysis (0.004), Technical Support (0.002), Microsoft Excel (0.002)	0.011
16	Delivery Driver And Technician	Rotech Healthcare	Physical Abilities (0.000), Lifting Ability (0.000), Caregiving (0.000), Patient Contact (0.000), Patient Transportation and Transfer (0.000), HAZMAT (0.000), Hazardous Materials Endorsement (0.000)	0
17	Vice President Underwriting	Morgan Stanley	Workflow Management (0.005), Written Communication (0.003), Detail-Oriented (0.002), Financial Analysis (0.001), Mortgage Underwriting (0.001), Staff Management (0.001)	0.002
18	Quality Assurance Engineer	Amazon	Computer Engineering (0.034), Software Development (0.027), User Interface (UI) Design (0.016), Software Quality Assurance (0.010), Black-box testing (0.009), Quality Assurance and Control (0.003), Consumer Electronics (0.002)	0.014
19	Sales Associate	GNC	Sales (0.001), Retail Industry Knowledge (0.000), Retail Sales (0.000), Basic Mathematics (0.000)	0
20	Dog And Cat Department Manager	Petco	Creativity (0.007), Leadership (0.003), Budgeting (0.001), Sales Goals (0.001), Retail Industry Knowledge (0.000), Physical Abilities (0.000), Inventory Management (0.000)	0.002

Table A3. Job Titles with the Highest Average AI-relatedness Measures

This table reports the job titles in Burning Glass with the highest average job-level AI measure ω_j^{AI} . We only include job titles that have at least 50 job postings and are matched to Compustat firms.

	Job Title	Avg. Continuous AI Measure
1	Artificial Intelligence Engineer	0.497
2	Senior Data Scientist - Machine Learning Engineer	0.394
3	Lead Machine Learning Scientist - Enterprise Products	0.369
4	AI Consultant	0.369
5	AI Senior Analyst	0.358
6	Machine Learning Engineer	0.315
7	Technician Architecture Delivery Senior Analyst AI	0.311
8	Artificial Intelligence Analyst	0.308
9	Software Engineer, Machine Learning	0.307
10	Artificial Intelligence Architect	0.303
11	Machine Learning Researcher	0.300
12	Computer Vision Engineer	0.293
13	Senior Machine Learning Engineer	0.286
14	Senior Machine Learning Scientist	0.281
15	Senior Software Engineer - Machine Learning	0.278
16	Senior Engineer II - Data Scientist	0.265
17	Senior Machine Learning Researcher	0.264
18	Artificial Intelligence Consultant	0.263
19	Computer Vision Scientist	0.256
20	Lead Machine Learning Researcher	0.255
21	Senior AI Engineer	0.248
22	Senior Applied Scientist	0.245
23	Senior Engineer - Machine Learning	0.243
24	Senior Risk Modeler	0.241
25	Data Scientist - Engineer	0.238
26	Artificial Intelligence Manager	0.237
27	Machine Learning Scientist	0.230
28	Applied Scientist	0.230
29	Software Engineer - Data Mining/Data Analysis/Machine Learning	0.229
30	Senior Associate, Data Scientist	0.223
31	Director, Data Scientist	0.222
32	Big Data Hadoop Consultant	0.214
33	Vice President- Data Analytics	0.211
34	Data Scientist Specialist	0.210
35	Applied Researcher	0.209
36	Data Scientist, Junior	0.205
37	Senior Staff Data Scientist	0.204
38	Principal Data Scientist	0.204
39	Director, Data Science	0.203
40	Research And Development Engineer - Data Mining/Data Analysis/Machine Learning	0.195
41	Manager, Data Scientist	0.192
42	Big Data Scientist	0.191
43	Architect - Relevance Infrastructure	0.191
44	Director Of Data Science	0.189
45	Senior Manager, Data Science	0.189
46	Data Science Specialist	0.188
47	Data Scientist II	0.188
48	Senior Data Science Engineer	0.187
49	Staff Data Scientist	0.186
50	Lead Data Scientist	0.186

Table A4. Firm-level Determinants of Job-postings-based AI Investments

This table reports the coefficients from regressions of cross-sectional changes in AI investments by U.S. public firms (in non-tech sectors) from 2010 to 2018 on the following ex-ante firm characteristics measured in 2010: log sales in column 1, cash/assets in column 2, R&D/sales in column 3, revenue TFP in column 4, log markup measured following [De Loecker et al. \(2020\)](#) in column 5, Tobin's Q in column 6, market leverage in column 7, return on assets (ROA) in column 8, and firm age in column 9. The dependent variable is the growth in the share of AI workers from 2010 to 2018 using the job postings data from Burning Glass. Regressions are weighted by the number of Burning Glass job postings in 2010. All specifications control for industry sector fixed effects. The dependent variable is normalized to have a mean of zero and a standard deviation of one. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Share of AI Workers, 2010–2018									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Log Sales 2010	0.156*** (0.032)									0.278*** (0.059)
Cash/Assets 2010		3.387** (1.376)								2.430*** (0.913)
R&D/Sales 2010			2.722 (2.230)							1.285 (2.017)
Revenue TFP 2010				0.541 (0.972)						-0.081 (0.494)
Log Markup 2010					0.449* (0.255)					0.314 (0.192)
Tobin's Q 2010						0.394* (0.216)				0.240 (0.207)
Market Leverage 2010							-1.102 (0.763)			-0.028 (0.368)
ROA 2010								2.814* (1.469)		0.592 (1.988)
Firm Age 2010									-0.005 (0.005)	-0.006* (0.003)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Adj R-Squared	0.184	0.261	0.167	0.148	0.168	0.237	0.158	0.174	0.150	0.395
Observations	1,082	1,082	1,082	1,082	1,082	1,082	1,082	1,082	1,082	1,082

Table A5. Summary Statistics

This table reports summary statistics for the sample of firms in our baseline regressions (including 1052 firms in the regressions with the resume-based measure and 935 firms in the regressions with the job-postings-based measure). All changes in variables are computed over 2010–2018. Changes in the numbers of trademarks and patents are measured as changes in $\log(1+\text{number})$ to take into account firms with zero trademarks or patents. We follow the methodology proposed by [Ganglmair et al. \(2021\)](#) to distinguish between product patents and process patents. The number of product and process patents are de-trended using the entire patent sample due to the truncation of the patent sample in recent years. The change in the number of trademarks (patents) is normalized by the change in the total number of trademarks (patents) from 2010 to 2018 to adjust for truncation in recent years. The change in the product mix is measured as the sum of annual changes from 2010 to 2018, where each annual change is the angle between the two word vectors indicating firms' product offerings in that year and the previous year following [Hoberg et al. \(2014\)](#). For each variable, we report the number of observations, the mean, the standard deviation, the median, and 1st, 5th, 10th, 25th, 75th, 90th, 95th, and 99th percentiles.

Variable Name	N	Mean	Std. Deviation	p1	p5	p10	p25	p50	p75	p90	p95	p99
Change in share of AI workers (Cognism)	1052	.0010	.0022	-.0018	-.00011	0	0	.00024	.0011	.0029	.0046	.014
Change in share of AI workers (Burning Glass)	935	.0046	.012	-.0033	0	0	0	0	.0034	.013	.026	.076
Change in log sales	1052	.45	.66	-1.2	-.61	-.24	.073	.39	.77	1.3	1.7	2.6
Change in log employment	1052	.3	.64	-1.8	-.67	-.35	-.036	.26	.61	1.0	1.4	2.2
Change in log market value	1010	.52	.72	-1.2	-.78	-.38	.1	.51	.94	1.4	1.7	2.6
Change in log number of trademarks	553	-.13	1.2	-3.0	-2.3	-1.8	-.92	0	.41	1.4	2.0	3.2
Change in log number of product patents	621	-.20	1.1	-2.9	-2.2	-1.8	-.94	.062	.85	.85	.85	1.5
Change in log number of process patents	621	-.067	.99	-3.0	-2.0	-1.5	-.63	.067	.76	.76	.76	1.5
Change in product mix	958	4.3	1.3	2.1	2.6	3.0	3.4	4.1	5.0	6.0	6.9	7.9
Change in log sales per worker	1052	.14	.39	-1.0	-.44	-.24	-.038	.12	.30	.56	.79	1.6
Change in revenue TFP	977	.0065	.36	-1.2	-.51	-.36	-.17	.0016	.17	.38	.55	1.2
Change in log COGS	1052	.38	.67	-1.6	-.80	-.34	.038	.35	.72	1.2	1.6	2.3
Change in log operating expense	1052	.42	.61	-1.3	-.57	-.22	.073	.37	.74	1.2	1.6	2.1
Employment in 2010 (thousands)	1052	23	55	.066	.21	.39	1.2	4.5	16	57	109	294
Sales in 2010 (millions)	1052	9104	24318	16	69	146	491	1601	5965	20732	50272	125805
Cash / Assets in 2010	1052	.16	.17	.00054	.0062	.014	.041	.11	.23	.41	.55	.70
Firm age in 2010	1052	25	18	1	4	6	12	19	37	59	60	60

Table A6. AI Investments and Stock Market Returns Using the Resume-based AI Measure

This table reports the coefficients from long-differences regressions of buy-and-hold returns from January 2010 to December 2018 on the contemporaneous firm-level changes in AI investments among U.S. public firms (in non-tech sectors). We consider four measures of returns: unadjusted holding period returns (columns 1 and 2), CAPM-adjusted returns (columns 3 and 4), FF3-adjusted returns (columns 5 and 6), and FF4-adjusted returns (columns 7 and 8). The main independent variable is the growth in the share of AI workers (based on the resume data) from 2010 to 2018, standardized to mean zero and standard deviation of one. Regressions are weighted by the number of Cognism resumes in 2010. All specifications control for industry sector fixed effects. Columns 2, 4, 6, and 8 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Buy-and-hold Return		CAPM-adjusted Return		FF3-adjusted Return		FF4-adjusted Return	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ Share AI Workers	0.822* (0.423)	0.646* (0.370)	0.287** (0.120)	0.227** (0.113)	0.187** (0.081)	0.180** (0.087)	0.212** (0.093)	0.192** (0.095)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y	N	Y
Adj R-Squared	0.276	0.396	0.256	0.357	0.200	0.300	0.191	0.290
Observations	799	799	799	799	799	799	799	799

Table A7. AI Investments and Firm Growth across Non-Tech Sectors Using the Resume-based AI Measure

This table reports the coefficients from long-differences regressions of firm growth from 2010 to 2018 on the contemporaneous changes in AI investments among U.S. public firms (in non-tech sectors), separately by broad industry sector. Columns 1 and 2 consider firms in the manufacturing sector (2-digit NAICS = 31, 32, 33), columns 3 and 4 consider firms in the wholesale and retail trade sectors (2-digit NAICS = 42, 44, 45), columns 5 and 6 look at firms in the finance sector (2-digit NAICS = 52), and columns 7 and 8 include firms in the other non-tech sectors (all 2-digit NAICS sectors, except those listed above and 51 and 54). The changes in the share of AI workers are based on the resume data and standardized to mean zero and standard deviation of one within each sample. We consider two measures of firm growth: changes in log sales in odd columns and changes in log employment in even columns. Regressions are weighted by the number of Cognism resumes in 2010. All regressions include industry sector fixed effects and the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Manufacturing		Wholesale & Retail		Finance		Other	
	Δ Log Sales	Δ Log Employment	Δ Log Sales	Δ Log Employment	Δ Log Sales	Δ Log Employment	Δ Log Sales	Δ Log Employment
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ Share AI Workers	0.134** (0.056)	0.098 (0.062)	0.364*** (0.054)	0.413*** (0.051)	0.273*** (0.087)	0.292*** (0.087)	0.127** (0.052)	0.065 (0.059)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y
Adj R-Squared	0.323	0.329	0.818	0.840	0.501	0.420	0.465	0.361
Observations	516	516	109	109	149	149	278	278

Table A8. AI Investments and Firm Growth in Tech Sectors Using the Resume-based AI Measure

This table reports the coefficients from long-differences regressions of firm growth from 2010 to 2018 on the contemporaneous changes in AI investments among U.S. public firms in tech sectors. Columns 1 and 2 consider firms in the information sector (2-digit NAICS = 51), and columns 3 and 4 consider firms in the professional and business services sector (2-digit NAICS = 54). The changes in the share of AI workers are based on the resume data and standardized to mean zero and standard deviation of one within each sample. We consider two measures of firm growth: changes in log sales in odd columns and changes in log employment in even columns. Regressions are weighted by the number of Cognism resumes in 2010. All regressions include industry sector fixed effects and the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Information		Prof. & Business Svcs	
	Δ Log Sales	Δ Log Employment	Δ Log Sales	Δ Log Employment
	(1)	(2)	(3)	(4)
Δ Share AI Workers	0.468*** (0.094)	0.381*** (0.072)	-0.058 (0.106)	0.112 (0.072)
Industry FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y
Adj R-Squared	0.636	0.602	0.535	0.365
Obs	130	130	56	56

Table A9. AI Investments and Firm Growth Excluding the Largest Firms

This table reports the coefficients from long-differences regressions of firm growth from 2010 to 2018 on the contemporaneous firm-level changes in AI investments among U.S. public firms (in non-tech sectors) excluding the largest firms. In Panel 1, the sample excludes firms in the top 1% of the size distribution in terms of 2010 sales; in Panel 2, the sample excludes firms in the top 5% of the size distribution in terms of 2010 sales. In each panel, we consider three measures of firm growth: changes in log sales (columns 1 and 2), changes in log employment (columns 3 and 4), and changes in log market value (columns 5 and 6). The dependent variables are measured as growth from 2010 to 2018. The main independent variable is the growth in the share of AI workers (based on the resume data) from 2010 to 2018, standardized to mean zero and standard deviation of one. Regressions are weighted by the number of Cognism resumes in 2010. All specifications control for industry sector fixed effects. Columns 2, 4, and 6 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel 1: Drop Top 1%

	Δ Log Sales		Δ Log Employment		Δ Log Market Value	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Share AI Workers	0.199*** (0.072)	0.193*** (0.068)	0.231** (0.100)	0.175** (0.086)	0.238** (0.096)	0.226** (0.089)
Industry FE	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y
Adj R-Squared	0.210	0.425	0.253	0.399	0.242	0.387
Observations	1,041	1,041	1,041	1,041	998	998

Panel 2: Drop Top 5%

	Δ Log Sales		Δ Log Employment		Δ Log Market Value	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Share AI Workers	0.182*** (0.068)	0.186*** (0.058)	0.206** (0.098)	0.162** (0.076)	0.220** (0.093)	0.237*** (0.085)
Industry FE	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y
Adj R-Squared	0.263	0.442	0.289	0.426	0.299	0.400
Observations	998	998	998	998	956	956

Table A10. AI Investments and Firm Growth: Long-differences Estimates Using the Job-postings-based AI Measure

This table reports the coefficients from long-differences regressions of firm growth from 2010 to 2018 on the contemporaneous firm-level changes in AI investments among U.S. public firms (in non-tech sectors). We consider three measures of firm growth: changes in log sales (columns 1 and 2), changes in log employment (columns 3 and 4), and changes in log market value (columns 5 and 6). The dependent variables are measured as growth from 2010 to 2018. The main independent variable is the growth in the share of AI job postings from 2010 to 2018, standardized to mean zero and standard deviation of one. Regressions are weighted by the number of Burning Glass job postings in 2010. All specifications control for industry sector fixed effects. Columns 2, 4, and 6 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of job postings), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log Sales		Δ Log Employment		Δ Log Market Value	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Share AI Workers	0.150*** (0.055)	0.158*** (0.042)	0.161** (0.076)	0.122** (0.050)	0.153* (0.082)	0.183*** (0.067)
Industry FE	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y
Adj R-Squared	0.273	0.433	0.320	0.512	0.364	0.474
Observations	935	935	935	935	902	902

Table A11. AI Investments and Firm Growth: Long-differences Estimates Using the Continuous Job-postings-based AI Measure

This table reports the coefficients from long-differences regressions of growth of U.S. public firms (in non-tech sectors) from 2010 to 2018 on the contemporaneous changes in the average job-level continuous AI measure across all Burning Glass job postings of each firm. The independent variable is the change in the firm-level average continuous AI-relatedness measures from 2010 to 2018, standradized to mean zero and standard deviation of one. Panel 1 presents results from the continuous measure based on all skills (ω_j^{AI}), and Panel 2 shows results from the continuous measure based on narrow AI skills ($\omega_j^{NarrowAI}$). The continuous measure based on narrow AI skills removes the impact of general programming or statistics skills not specific to AI (e.g., Python or linear regression). See Section 3.1 for detailed descriptions of these measures. We consider three measures of firm growth: changes in log sales (columns 1 and 2), changes in log employment (columns 3 and 4), and changes in log market value (columns 5 and 6). Regressions are weighted by the number of Burning Glass job postings in 2010. All specifications control for industry sector fixed effects. Columns 2, 4, and 6 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, log number of job postings), log industry wage, as well as characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel 1: Continuous All-skill AI Measure

	Δ Log Sales		Δ Log Employment		Δ Log Market Value	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Share AI Workers	0.143** (0.069)	0.163*** (0.054)	0.144 (0.093)	0.116* (0.068)	0.144 (0.094)	0.183** (0.079)
Industry FE	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y
Adj R-Squared	0.254	0.422	0.306	0.506	0.352	0.465
Observations	935	935	935	935	902	902

Panel 2: Continuous Narrow AI measure

	Δ Log Sales		Δ Log Employment		Δ Log Market Value	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Share AI Workers	0.147*** (0.057)	0.165*** (0.043)	0.154* (0.079)	0.124** (0.052)	0.148* (0.084)	0.188*** (0.069)
Industry FE	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y
Adj R-Squared	0.271	0.435	0.317	0.511	0.361	0.474
Observations	935	935	935	935	902	902

Table A12. AI Investments and Firm Growth: Long-differences Estimates Using Alternative Cutoffs of the Job-postings-based AI Measure

This table reports the coefficients from long-differences regressions of firm growth from 2010 to 2018 on the contemporaneous changes in the share of AI job postings of U.S. public firms (in non-tech sectors). AI job postings are defined as job postings with continuous job-level measure ω_j^{AI} above 0.05 in Panel 1, and job postings with continuous job-level measure ω_j^{AI} above 0.15 in Panel 2. See Section 3.1 for the detailed description of the methodology. The independent variable is standardized to mean zero and standard deviation of one. We consider three measures of firm growth: changes in log sales (columns 1 and 2), changes in log employment (columns 3 and 4), and changes in log market value (columns 5 and 6). Regressions are weighted by the number of Burning Glass job postings in 2010. All regressions control for industry sector fixed effects. Columns 2, 4, and 6 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number job postings), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel 1: Cutoff = 0.05

	<u>Δ Log Sales</u>		<u>Δ Log Employment</u>		<u>Δ Log Market Value</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Share AI Workers	0.142** (0.058)	0.164*** (0.040)	0.153* (0.080)	0.130*** (0.050)	0.141* (0.085)	0.184*** (0.067)
Industry FE	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y
Adj R-Squared	0.268	0.434	0.317	0.513	0.359	0.472
Observations	935	935	935	935	902	902

Panel 2: Cutoff = 0.15

	<u>Δ Log Sales</u>		<u>Δ Log Employment</u>		<u>Δ Log Market Value</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Share AI Workers	0.146** (0.065)	0.142*** (0.050)	0.154* (0.085)	0.105* (0.055)	0.154* (0.091)	0.173** (0.077)
Industry FE	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y
Adj R-Squared	0.261	0.419	0.311	0.507	0.359	0.467
Observations	935	935	935	935	902	902

Table A13. AI Investments and Firm Growth for Top AI-investing Firms in the Distribution of Resume-based AI Investments

This table reports the coefficients from long-differences regressions of firm growth from 2010 to 2018 on measures of contemporaneous AI investments among U.S. public firms (in non-tech sectors). We consider three measures of firm growth: changes in log sales (columns 1 and 2), changes in log employment (columns 3 and 4), and changes in log market value (columns 5 and 6). The independent variable is a dummy variable for firms having changes in the (resume-based) share of AI workers in the top 25% of the distribution (in Panel 1) or in the top 10% of the distribution (in Panel 2). Regressions are weighted by the number of Cognism resumes in 2010. All specifications control for industry sector fixed effects. Columns 2, 4, and 6 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel 1: Top 25% of AI-investing Firms

	Δ Log Sales		Δ Log Employment		Δ Log Market Value	
	(1)	(2)	(3)	(4)	(5)	(6)
Top 25% in AI Investment	0.302** (0.137)	0.307*** (0.092)	0.354* (0.186)	0.282** (0.115)	0.276 (0.192)	0.197* (0.111)
Industry FE	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y
Adj R-Squared	0.153	0.384	0.162	0.379	0.163	0.311
Observations	1,052	1,052	1,052	1,052	1,009	1,009

Panel 2: Top 10% of AI-investing Firms

	Δ Log Sales		Δ Log Employment		Δ Log Market Value	
	(1)	(2)	(3)	(4)	(5)	(6)
Top 10% in AI Investment	0.667*** (0.182)	0.535*** (0.094)	0.812*** (0.241)	0.577*** (0.135)	0.822*** (0.236)	0.698*** (0.143)
Industry FE	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y
Adj R-Squared	0.214	0.404	0.236	0.403	0.249	0.360
Observations	1,052	1,052	1,052	1,052	1,009	1,009

Table A14. AI Investments (Including External AI Software) and Firm Growth Using the Resume-based AI Measure

This table reports the coefficients from long-differences regressions of firm growth from 2010 to 2018 on the contemporaneous firm-level changes in AI investments among U.S. public firms (in non-tech sectors). The main independent variable is the growth in the share of AI-utilizing workers (including both workers with AI skills and workers with jobs referencing external AI software, such as IBM Watson) from 2010 to 2018, standardized to mean zero and standard deviation of one. We consider three measures of firm growth: changes in log sales (columns 1 and 2), changes in log employment (columns 3 and 4), and changes in log market value (columns 5 and 6). Regressions are weighted by the number of Cognism resumes in 2010. All specifications control for industry sector fixed effects. Columns 2, 4, and 6 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log Sales		Δ Log Employment		Δ Log Market Value	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Share AI Workers	0.242*** (0.056)	0.224*** (0.052)	0.289*** (0.081)	0.233*** (0.066)	0.287*** (0.072)	0.271*** (0.063)
Industry FE	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y
Adj R-Squared	0.245	0.437	0.268	0.430	0.272	0.388
Observations	1,023	1,023	1,023	1,023	993	993

Table A15. AI Investments and Firm Growth : Controlling for Other Technologies

This table reports the coefficients from long-differences regressions of firm growth from 2010 to 2018 on the contemporaneous changes in AI investments among U.S. public firms (in non-tech sectors), controlling for investments in other (non-AI-specific) technologies. Columns 1 and 2 control for the 2010-2018 change in the firm's share of non-AI IT jobs, columns 3 and 4 control for the change in the share of robot-related jobs, columns 5 and 6 control for the change in the share of non-AI data-related jobs, and columns 7 and 8 control for the changes in the share of non-AI data analysis jobs. An IT job is defined as a job for which at least 10% of the required skills are in the "Information Technology" skill cluster; a robot-related job is a job with a robot relatedness score (constructed with the same methodology as the AI-relatedness score but using the core skill of "Robotics") above 0.1; a data-related job is a job with at least 10% of required skills in data-related skill clusters; a data analysis job is a job with at least 10% of required skills in the "Analysis" skill cluster. All measures are standardized to mean zero and standard deviation of one. Panel 1 considers the resume-based measure of the share of AI workers, while Panel 2 looks at the job-posting-based measure. Regressions are weighted by the number of Cognism resumes in 2010 in Panel 1 and the number of Burning Glass job postings in 2010 in Panel 2. All regressions control for industry sector fixed effects and the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of jobs—resumes in Panel 1 and job postings in Panel 2), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel 1: AI measure from resume data								
	Δ Log Sales	Δ Log Employment	Δ Log Sales	Δ Log Employment	Δ Log Sales	Δ Log Employment	Δ Log Sales	Δ Log Employment
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ Share AI Workers	0.197*** (0.063)	0.182** (0.082)	0.197*** (0.074)	0.183** (0.093)	0.181*** (0.059)	0.166** (0.076)	0.185*** (0.063)	0.171** (0.080)
Δ Share Non-AI IT Workers	0.140*** (0.048)	0.124*** (0.047)						
Δ Share Robot Workers			-0.013 (0.034)	-0.018 (0.044)				
Δ Share Non-AI Data Workers					0.134*** (0.034)	0.139*** (0.036)		
Δ Share Non-AI Data Analysis Workers							0.076*** (0.029)	0.078** (0.034)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y
Adj R-Squared	0.450	0.423	0.427	0.410	0.457	0.435	0.437	0.417
Observations	970	970	970	970	970	970	970	970
Panel 2: AI measure from job postings data								
	Δ Log Sales	Δ Log Employment	Δ Log Sales	Δ Log Employment	Δ Log Sales	Δ Log Employment	Δ Log Sales	Δ Log Employment
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ Share AI Workers	0.148*** (0.041)	0.110** (0.049)	0.143*** (0.037)	0.117** (0.045)	0.152*** (0.040)	0.113** (0.046)	0.157*** (0.041)	0.121** (0.048)
Δ Share Non-AI IT Workers	0.152*** (0.053)	0.200*** (0.056)						
Δ Share Robot Workers			0.088** (0.041)	0.035 (0.049)				
Δ Share Non-AI Data Workers					0.046 (0.069)	0.082 (0.075)		
Δ Share Non-AI Data Analysis Workers							0.012 (0.055)	0.010 (0.053)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y
Adj R-Squared	0.455	0.530	0.440	0.511	0.434	0.513	0.432	0.511
Observations	935	935	935	935	935	935	935	935

Table A16. Resume-based AI Investments and Firm Growth: Controlling for Detailed Industry Fixed Effects

This table reports the coefficients from long-differences regressions of firm growth from 2010 to 2018 on the contemporaneous firm-level changes in AI investments among U.S. public firms (in non-tech sectors), with detailed industry fixed effects. We consider three measures of growth: changes in log sales (columns 1–4), changes in log employment (columns 5–8), and changes in log market value (columns 9–12). The main independent variable is the growth in the share of AI workers (based on the resume data) from 2010 to 2018, standardized to mean zero and standard deviation of one. Regressions are weighted by the number of Cognism resumes in 2010. All specifications include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Columns 1, 5, and 9 include the baseline specification with the industry sector fixed effects; columns 2, 6, and 10 control for 3-digit NAICS fixed effects; columns 3, 7, and 11 control for 4-digit NAICS fixed effects; columns 4, 8, and 12 control for 5-digit NAICS fixed effects. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log Sales				Δ Log Employment				Δ Log Market Value			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Δ Share AI Workers	0.195*** (0.069)	0.180*** (0.068)	0.195** (0.094)	0.208** (0.096)	0.181** (0.086)	0.150* (0.082)	0.180 (0.110)	0.183* (0.108)	0.223** (0.086)	0.182** (0.079)	0.199* (0.101)	0.208** (0.100)
NAICS2 FE	Y	N	N	N	Y	N	N	N	Y	N	N	N
NAICS3 FE	N	Y	N	N	N	Y	N	N	N	Y	N	N
NAICS4 FE	N	N	Y	N	N	N	Y	N	N	N	Y	N
NAICS5 FE	N	N	N	Y	N	N	N	Y	N	N	N	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Adj R-Squared	0.422	0.497	0.521	0.531	0.405	0.475	0.552	0.563	0.364	0.497	0.574	0.609
Observations	1,052	1,047	1,011	942	1,052	1,047	1,011	942	1,009	1,005	967	895

Table A17. Resume-based AI Investments and Firm Growth: Controlling for Industry- and Firm-level Pre-trends

This table reports the coefficients from long-differences regressions of firm growth from 2010 to 2018 on the contemporaneous firm-level changes in AI investments among U.S. public firms (in non-tech sectors). In this table, we control for past industry and firm growth, which helps address the concern that AI-investing firms might already be on higher growth trajectories prior to AI investments. We consider three measures of firm growth: changes in log sales (columns 1 and 2), changes in log employment (columns 3 and 4), and changes in log market value (columns 5 and 6). The main independent variable is the growth in the share of AI workers (based on the resume data) from 2010 to 2018, standardized to mean zero and standard deviation of one. Regressions are weighted by the number of Cognism resumes in 2010. All specifications control for industry sector fixed effects and include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Additionally, columns 1, 3, and 5 control for industry-level growth in sales and employment from 2000 to 2008 (at the 5-digit NAICS level), and columns 2, 4, and 6 control for firm-level growth in sales and employment from 2000 to 2008. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log Sales		Δ Log Employment		Δ Log Market Value	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Share AI Workers	0.182*** (0.070)	0.174*** (0.056)	0.184** (0.089)	0.173** (0.074)	0.197*** (0.076)	0.192*** (0.068)
Industry FE	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y
Industry pre-trend	Y	N	Y	N	Y	N
Firm pre-trend	N	Y	N	Y	N	Y
Observations	796	793	796	793	770	767

Table A18. Resume-based AI Investments and Firm Growth: Controlling for State FE and Tobin's q

This table reports the coefficients from long-differences regressions of firm growth from 2010 to 2018 on the contemporaneous firm-level changes in AI investments among U.S. public firms (in non-tech sectors), with additional controls for state fixed effects and Tobin's q. We consider three measures of growth: changes in log sales (columns 1 and 2), changes in log employment (columns 3 and 4), and changes in log market value (columns 5 and 6). The main independent variable is the growth in the share of AI workers (based on the resume data) from 2010 to 2018, standardized to mean zero and standard deviation of one. Regressions are weighted by the number of Cognism resumes in 2010. All specifications control for industry sector fixed effects, state fixed effects, and the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Columns 2, 4, and 6 also control for Tobin's Q in 2010, defined as market value of assets divided by book value of assets. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log Sales		Δ Log Employment		Δ Log Market Value	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Share AI Workers	0.177*** (0.060)	0.156*** (0.044)	0.180** (0.069)	0.158*** (0.053)	0.194*** (0.070)	0.170*** (0.053)
Tobin's Q 2010		0.225*** (0.065)		0.233*** (0.063)		0.245*** (0.076)
Industry FE	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y	Y
Adj R-Squared	0.493	0.554	0.510	0.564	0.425	0.474
Observations	1,045	1,003	1,045	1,003	1,000	999

Table A19. Past Resume-based AI Investments and Future Firm Growth

This table reports the coefficients from a predictive regression of firm growth during the later part of our sample (2015–2020) on growth in AI investments during the earlier part of the sample (2010–2015) among U.S. public firms (in non-tech sectors). The dependent variables are changes in log sales in columns 1 and 2, and changes in log employment in columns 3 and 4. The main independent variable is the growth in the share of AI workers (based on the resume data) from 2010 to 2015, standardized to mean zero and standard deviation of one. Regressions are weighted by the number of Cognism resumes in 2010. All specifications control for industry sector fixed effects. Columns 2 and 4 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log Sales		Δ Log Employment	
	(1)	(2)	(3)	(4)
Δ Share AI Workers	0.128** (0.053)	0.082** (0.039)	0.149* (0.079)	0.067 (0.055)
Industry FE	Y	Y	Y	Y
Controls	N	Y	N	Y
Adj R-Squared	0.250	0.367	0.211	0.357
Observations	990	990	982	982

Table A20. IV Estimates of AI Investments and Firm Growth Using Alternative CS Control

This table estimates the relationship between AI investments and firm growth from 2010 to 2018 for U.S. public firms (in non-tech sectors), where firm AI investments are instrumented with ex-ante firm-level exposure to AI-skilled graduates from AI-strong universities. The independent variable is the change in the share of AI workers from 2010 to 2018 based on the resume data. Regressions are weighted by the number of Cognism resumes in 2010 (see the definition of the instrument in Section 4.3 and the details of instrument construction in Appendix A). The independent variable and the instrument are standardized to mean zero and standard deviation of one. We consider changes in log sales in columns 1 to 4, log employment in columns 5 to 8, and log market value in columns 9 to 12. All specifications control for industry sector fixed effects and ex-ante exposure to universities that are strong in computer science research as well as top 10 universities. The ranking of CS departments is from csrankings.org and we excluded conferences in AI areas (including AI, computer vision, machine learning, and natural language processing) and limit the time period to 2005–2009. Columns 2–4, 6–8, and 10–12 also control for the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Columns 3–4, 7–8, and 11–12 additionally control for firm-level changes in log sales and log employment from 2000 to 2008. Columns 4, 8, and 12 add state fixed effects. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. The first-stage F-statistics of the instrument are reported for all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log Sales				Δ Log Employment				Δ Log Market Value			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Δ Share AI Workers	0.360*** (0.101)	0.510*** (0.125)	0.467*** (0.137)	0.317** (0.152)	0.466*** (0.154)	0.712*** (0.224)	0.584*** (0.194)	0.330** (0.163)	0.449*** (0.129)	0.529*** (0.160)	0.470*** (0.172)	0.360* (0.184)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CS Control	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Baseline Controls	N	Y	Y	Y	N	Y	Y	Y	N	Y	Y	Y
Control Pre-trend	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y
State FE	N	N	N	Y	N	N	N	Y	N	N	N	Y
F Statistic	12.7	19.3	15.7	18.7	12.7	19.3	15.7	18.7	12.8	19.5	15.5	19.3
Observations	1,001	1,001	777	773	1,001	1,001	777	773	963	963	753	746

Table A21. IV Estimates of AI Investments and Firm Growth Using AI Hubs Before 2005

This table estimates the relationship between AI investments and firm growth from 2010 to 2018 for U.S. public firms (in non-tech sectors), where firm AI investments are instrumented with ex-ante firm-level exposure to AI-skilled graduates from AI-strong universities. We define a university as an AI-strong university if it satisfies one of the following two criteria in at least one year between 2000 and 2004: (i) the number of AI researchers is in the top 5% of the distribution across all universities in a given year; (ii) the number of AI researchers is in the top 10% of the distribution, and the share of AI researchers is in the top 5% of the distribution across all universities in a given year (see the definition of the instrument in Section 4.3 and the details of instrument construction in Appendix A). The independent variable is the change in the share of AI workers from 2010 to 2018 based on the resume data. Regressions are weighted by the number of Cognism resumes in 2010. The independent variable and the instrument are standardized to mean zero and standard deviation of one. We consider changes in log sales in columns 1 to 4, log employment in columns 5 to 8, and log market value in columns 9 to 12. All specifications control for industry sector fixed effects and ex-ante exposure to universities that are strong in computer science research as well as top 10 universities. Columns 2–4, 6–8, and 10–12 also control for the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Columns 3–4, 7–8, and 11–12 additionally control for firm-level changes in log sales and log employment from 2000 to 2008. Columns 4, 8, and 12 add state fixed effects. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. The first-stage F-statistics of the instrument are reported for all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log Sales				Δ Log Employment				Δ Log Market Value			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Δ Share AI Workers	0.361*** (0.109)	0.467*** (0.130)	0.419*** (0.141)	0.288* (0.168)	0.444*** (0.166)	0.646*** (0.220)	0.500*** (0.187)	0.275 (0.180)	0.450*** (0.134)	0.494*** (0.172)	0.423** (0.183)	0.337* (0.203)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CS Control	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Baseline Controls	N	Y	Y	Y	N	Y	Y	Y	N	Y	Y	Y
Control Pre-trend	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y
State FE	N	N	N	Y	N	N	N	Y	N	N	N	Y
F Statistic	12.1	18.5	12.2	13.4	12.1	18.5	12.2	13.4	12.2	19.2	12.4	14.3
Observations	1,001	1,001	777	773	1,001	1,001	777	773	963	963	753	746

Table A22. IV Estimates of AI Investments and Firm Growth Excluding Prominent Universities

This table estimates the relationship between AI investments and firm growth from 2010 to 2018 for U.S. public firms (in non-tech sectors), where firm AI investments are instrumented with ex-ante firm-level exposure to AI-skilled graduates from AI-strong universities. In Panel 1, we exclude Stanford University from the construction of the instrument; in Panel 2, we exclude Carnegie Mellon University from the construction of the instrument; in Panel 3, we exclude University of Illinois Urbana-Champaign from the construction of the instrument (see the definition of the instrument in Section 4.3 and the details of instrument construction in Appendix A). The independent variable, dependent variables, and controls are the same as Table 5. Regressions are weighted by the number of Cognism resumes in 2010. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. The first-stage F-statistics of the instrument are reported for all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel 1: Exclude Stanford

	Δ Log Sales				Δ Log Employment				Δ Log Market Value			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Δ Share AI Workers	0.356*** (0.101)	0.498*** (0.124)	0.461*** (0.135)	0.308** (0.152)	0.465*** (0.154)	0.702*** (0.219)	0.581*** (0.191)	0.326** (0.164)	0.444*** (0.130)	0.517*** (0.159)	0.465*** (0.170)	0.354* (0.184)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CS Control	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Baseline Controls	N	Y	Y	Y	N	Y	Y	Y	N	Y	Y	Y
Control Pre-trend	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y
State FE	N	N	N	Y	N	N	N	Y	N	N	N	Y
F Statistic	12.4	19.4	15.8	17.8	12.4	19.4	15.8	17.8	12.5	19.6	15.6	18.4
Observations	1,001	1,001	777	773	1,001	1,001	777	773	963	963	753	746

Panel 2: Exclude CMU

	Δ Log Sales				Δ Log Employment				Δ Log Market Value			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Δ Share AI Workers	0.365*** (0.100)	0.523*** (0.127)	0.481*** (0.141)	0.326** (0.153)	0.476*** (0.156)	0.736*** (0.233)	0.603*** (0.201)	0.342** (0.167)	0.458*** (0.129)	0.548*** (0.163)	0.489*** (0.176)	0.378** (0.188)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CS Control	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Baseline Controls	N	Y	Y	Y	N	Y	Y	Y	N	Y	Y	Y
Control Pre-trend	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y
State FE	N	N	N	Y	N	N	N	Y	N	N	N	Y
F Statistic	12.3	18.9	15.6	18.0	12.3	18.9	15.6	18.0	12.5	19.1	15.4	18.6
Observations	1,001	1,001	777	773	1,001	1,001	777	773	963	963	753	746

Panel 3: Exclude UIUC

	Δ Log Sales				Δ Log Employment				Δ Log Market Value			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Δ Share AI Workers	0.364*** (0.100)	0.501*** (0.121)	0.463*** (0.133)	0.361** (0.165)	0.469*** (0.153)	0.698*** (0.218)	0.584*** (0.190)	0.370** (0.177)	0.460*** (0.125)	0.543*** (0.152)	0.486*** (0.168)	0.439** (0.199)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CS Control	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Baseline Controls	N	Y	Y	Y	N	Y	Y	Y	N	Y	Y	Y
Control Pre-trend	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y
State FE	N	N	N	Y	N	N	N	Y	N	N	N	Y
F Statistic	13.0	19.4	16.6	20.8	13.0	19.4	16.6	20.8	13.1	19.6	16.4	21.4
Observations	1,001	1,001	777	773	1,001	1,001	777	773	963	963	753	746

Table A23. IV Estimates of AI Investments and Firm Growth Using Non-Stem Workers

This table estimates the relationship between AI investments and firm growth from 2010 to 2018 for U.S. public firms (in non-tech sectors), where firm AI investments are instrumented with ex-ante firm-level exposure to AI-skilled graduates from AI-strong universities. The instrument is defined as the share of non-STEM workers in the firm in 2010 who graduated from AI-strong universities. The independent variable is the change in the share of AI workers from 2010 to 2018 based on the resume data. Regressions are weighted by the number of Cognism resumes in 2010. The independent variable and the instrument are standardized to mean zero and standard deviation of one. We consider changes in log sales in columns 1 to 4, log employment in columns 5 to 8, and log market value in columns 9 to 12. All specifications control for industry sector fixed effects and ex-ante exposure to universities that are strong in computer science research as well as top 10 universities. Columns 2–4, 6–8, and 10–12 also control for the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Columns 3–4, 7–8, and 11–12 additionally control for firm-level changes in log sales and log employment from 2000 to 2008. Columns 4, 8, and 12 add state fixed effects. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. The first-stage F-statistics of the instrument are reported for all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log Sales				Δ Log Employment				Δ Log Market Value			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Δ Share AI Workers	0.365*** (0.111)	0.358** (0.138)	0.316*** (0.120)	0.177* (0.095)	0.430*** (0.133)	0.389*** (0.147)	0.350*** (0.129)	0.119 (0.097)	0.433*** (0.145)	0.416** (0.170)	0.332** (0.147)	0.214* (0.122)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CS Control	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Baseline Controls	N	Y	Y	Y	N	Y	Y	Y	N	Y	Y	Y
Control Pre-trend	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y
State FE	N	N	N	Y	N	N	N	Y	N	N	N	Y
F Statistic	11.7	17.3	14.5	9.6	11.7	17.3	14.5	9.6	11.9	17.8	15.0	9.6
Observations	1,001	1,001	777	773	1,001	1,001	777	773	963	963	753	746

Table A24. AI Investments and Product Innovation Using the Job-postings-based AI Measure

This table reports the coefficients from long-differences regressions of the changes in measures of product innovation from 2010 to 2018 on the contemporaneous changes in AI investments by U.S. public firms (in non-tech sectors). The dependent variables are the change in $\log(1+\text{number of trademarks})$ in columns 1 and 2; the change in $\log(1+\text{number of product patents})$ in columns 3 and 4; and the change in the product mix in columns 5 and 6. Product patents are patents with over 50% of the claims being product claims, following the categorization in [Ganglmair et al. \(2021\)](#). The change in the product mix is measured as the sum of annual changes from 2010 to 2018, where each annual change is the angle between the two word vectors indicating firms' product offerings in that year and the previous year (the word vectors are constructed as in [Hoberg et al. \(2014\)](#)). The main independent variable is the job-postings-based measure of the growth in the share of AI workers from 2010 to 2018, which is standardized to mean zero and standard deviation of one. Regressions are weighted by the number of Burning Glass job postings in 2010. All specifications control for industry sector fixed effects. Columns 2, 4, and 6 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of job postings), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Columns 1 and 2 control for the log number of trademarks before 2010, and columns 3 and 4 control for the log number of patents before 2010. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log Number of Trademarks		Δ Log Number of Product Patents		Change in Product Mix	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Share AI Workers	0.132** (0.057)	0.051 (0.070)	0.169*** (0.029)	0.191*** (0.046)	0.109*** (0.033)	0.023 (0.041)
Industry FE	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y
Observations	505	505	560	560	860	860

Table A25. AI Investments and Operating Costs Using the Job-postings-based AI Measure

This table reports the coefficients from long-differences regressions of changes in firm operating costs and firm productivity from 2010 to 2018 on contemporaneous changes in AI investments by U.S. public firms (in non-tech sectors). The main independent variable is the change in the share of AI workers (based on the job postings data) from 2010 to 2018, standardized to mean zero and standard deviation of one. We look at two measures of operating costs: log COGS in columns 1 and 2 and log operating expenses in columns 3 and 4. We consider two measures of productivity: log sales per worker (columns 5–6) and revenue TFP (columns 7–8). Revenue TFP is the residual from regressing log revenue on log employment and log capital (constructed using the perpetual inventory method), with separate regressions for each industry sector. In columns 9 and 10, the dependent variable is the change in $\log(1 + \text{number of process patents})$, where process patents are patents with over 50% of the claims being process claims, following the categorization in [Ganglmair et al. \(2021\)](#). Regressions are weighted by the number of Burning Glass job postings in 2010. All specifications control for industry sector fixed effects. Columns 2, 4, 6, and 8 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of job postings), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Columns 9 and 10 also control for the log number of patents before 2010. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log COGS		Δ Log Operating Expense		Δ Log Sales per Worker		Δ Revenue TFP		Δ Log Number of Process Patents	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Δ Share AI Workers	0.153*** (0.039)	0.143*** (0.035)	0.148*** (0.051)	0.147*** (0.039)	-0.011 (0.026)	0.036 (0.022)	-0.018 (0.042)	-0.001 (0.031)	0.031 (0.048)	0.071 (0.063)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y	N	Y	N	Y
Adj R-Squared	0.246	0.384	0.251	0.405	0.429	0.618	0.287	0.401	0.732	0.749
Observations	935	935	935	935	935	935	874	874	560	560

Table A26. AI Investments and Operating Costs: IV Estimates Using the Resume-based AI Measure

This table estimates the relationship between AI investments and operating costs and productivity from 2010 to 2018 for U.S. public firms (in non-tech sectors), where firm AI investments are instrumented with ex-ante firm-level exposure to AI-skilled graduates from AI-strong universities. The independent variable is the change in the share of AI workers from 2010 to 2018 based on the resume data. Regressions are weighted by the number of Cognism resumes in 2010 (see the definition of the instrument in Section 4.3 and the details of instrument construction in Appendix A). The independent variable and the instrument are standardized to mean zero and standard deviation of one. We look at two measures of operating costs: log COGS in columns 1–4 and log operating expenses in columns 5–8. We consider two measures of productivity: log sales per worker (columns 9–12) and revenue TFP (columns 13–16). Revenue TFP is the residual from regressing log revenue on log employment and log capital (constructed using the perpetual inventory method), with separate regressions for each industry sector. In columns 17–20, the dependent variable is the change in log(1+number of process patents), where process patents are patents with over 50% of the claims being process claims, following the categorization in Ganglmair et al. (2021). All specifications control for industry sector fixed effects and ex-ante exposure to universities that are strong in computer science research as well as top 10 universities. Columns 2–4, 6–8, 10–12, 14–16, and 18–20 also control for the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Columns 3–4, 7–8, 11–12, 15–16, and 19–20 additionally control for firm-level changes in log sales and log employment from 2000 to 2008. Columns 4, 8, 12, 16, and 20 add state fixed effects. Columns 17–20 also control for the log number of patents before 2010. Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. The first-stage F-statistics of the instrument are reported for all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log COGS				Δ Log Operating Expense				Δ Log Sales per Worker				Δ Revenue TFP				Δ Log Number of Process Patents			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
Δ Share AI Workers	0.357*** (0.090)	0.497*** (0.126)	0.464*** (0.131)	0.366** (0.163)	0.361*** (0.099)	0.518*** (0.129)	0.480*** (0.138)	0.315** (0.148)	-0.106 (0.083)	-0.202 (0.135)	-0.117 (0.082)	-0.013 (0.042)	-0.094* (0.057)	-0.134* (0.070)	-0.087 (0.055)	-0.042 (0.047)	-0.353 (0.279)	-0.168 (0.272)	-0.041 (0.248)	0.114 (0.144)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CS Control	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Baseline Controls	N	Y	Y	Y	N	Y	Y	Y	N	Y	Y	Y	N	Y	Y	Y	N	Y	Y	Y
Control Pre-trend	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y
State FE	N	N	N	Y	N	N	N	Y	N	N	N	Y	N	N	N	Y	N	N	N	Y
F Statistic	12.7	19.3	15.7	18.7	12.7	19.3	15.7	18.7	12.7	19.3	15.7	18.7	13.1	19.6	17.7	21.3	11.5	23.8	25.4	34.6
Observations	1,001	1,001	777	773	1,001	1,001	777	773	1,001	1,001	777	773	929	929	742	738	586	586	479	469

Table A27. Resume-based AI Investments and Productivity of Early Adopters

This table reports the coefficients from long-differences regressions of changes in firm productivity from 2010 to 2018 on changes in AI investments by U.S. public firms (in non-tech sectors) from 2010 to 2014. We consider two measures of productivity: log sales per worker in columns 1 and 2 and revenue TFP in columns 3 and 4. Revenue TFP is the residual from regressing log revenue on log employment and log capital (constructed using the perpetual inventory method), with separate regressions for each industry sector. The main independent variable is growth in the share of AI workers based on the resume data from 2010 to 2014. All independent variables are standardized to mean zero and standard deviation of one. Regressions are weighted by the number of Cognism resumes in 2010. All specifications control for industry sector fixed effects. Columns 2 and 4 also include the baseline controls all measured as of 2010: firm-level characteristics (log sales, cash/assets, firm age, and log number of resumes), log industry wage, and characteristics of the commuting zones where the firms are located (the share of workers in IT-related occupations, the share of college-educated workers, log average wage, the share of foreign-born workers, the share of routine workers, the share of workers in finance and manufacturing industries, and the share of female workers). Standard errors are clustered at the 5-digit NAICS industry level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log Sales per Worker		Δ Revenue TFP	
	(1)	(2)	(3)	(4)
Δ Share AI Workers 2010-2014	-0.026 (0.030)	0.025 (0.023)	-0.041 (0.053)	0.006 (0.038)
Industry FE	Y	Y	Y	Y
Controls	N	Y	N	Y
Adj R-Squared	0.170	0.322	0.211	0.358
Observations	1,033	1,033	961	961

Table A28. Resume-based AI Investments and Industry Growth within a Balanced Panel of Firms

This table reports the coefficients from industry-level long-differences regressions of the changes in total industry sales and employment on contemporaneous changes in AI investments for a balanced panel of firms existing in both 2010 and 2018. Each observation is a 5-digit NAICS industry, and we exclude tech sectors. The independent variable is the change in the share of AI workers based on the resume data from 2010 to 2018, standardized to mean zero and standard deviation of one. Regressions are weighted by the total (industry-level) number of Cognism resumes in 2010. The dependent variables are the changes, from 2010 to 2018, in log total sales (not including entrants and exits between 2010 and 2018) in columns 1 and 2 and in log total employment in columns 3 and 4. All measures are calculated using Compustat firms that exist at the beginning (2010) and the end (2018) of our sample. All specifications control for industry sector fixed effects. Regressions in columns 2 and 4 also include industry-level controls for log total employment, log total sales, and log average wage in 2010. Standard errors are robust against heteroskedasticity and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Sales		Employment	
	(1)	(2)	(3)	(4)
Δ Share AI Workers	0.172*** (0.058)	0.188*** (0.045)	0.195*** (0.074)	0.211*** (0.062)
Industry FE	Y	Y	Y	Y
Controls	N	Y	N	Y
Observations	275	275	275	275

Table A29. AI Investments and Changes in Industry Growth and Concentration Using the Job-postings-based AI Measure

This table reports the coefficients from industry-level long-differences regressions of the changes in industry sales, employment, and concentration on contemporaneous changes in industry-level AI investments. All industry-level variables are calculated for all firms in Compustat (regardless of whether they are in our main regression sample in Table 3 or not). Each observation is a 5-digit NAICS industry, and (as in our main analysis) we exclude tech sectors. The independent variable is the change in the share of AI workers (based on the job postings data) from 2010 to 2018, standardized to mean zero and standard deviation of one. Regressions are weighted by the total (industry-level) number of Burning Glass job postings in 2010. The dependent variables are the changes, from 2010 to 2018, in log total sales in columns 1 and 2, log total employment in columns 3 and 4, the Herfindahl-Hirschman Index (HHI) in columns 5 and 6, and the market share of the top firm in an industry in columns 7 and 8. All specifications control for industry sector fixed effects. Regressions in columns 2, 4, 6, and 8 also include industry-level controls for log total employment, log total sales, and log average wage in 2010. Standard errors are robust against heteroskedasticity and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Log Sales		Log Employment		HHI		Top Firm Market Share	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ Share AI Workers	0.152** (0.072)	0.129** (0.062)	0.152 (0.113)	0.131 (0.102)	0.037*** (0.006)	0.040*** (0.006)	0.041*** (0.006)	0.045*** (0.006)
Industry Sector FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y	N	Y
Observations	261	261	261	261	261	261	261	261

Table A30. AI Investments and Changes in Industry Growth and Concentration Using Census Data

This table reports the coefficients from industry-level long-differences regressions of the changes in industry sales, employment, and concentration from 2012 to 2017 on contemporaneous changes in industry-level AI investments. All industry-level dependent variables are from the Economic Census. Each observation is a 5-digit NAICS industry, and (as in our main analysis) we exclude tech sectors. The independent variable is the change in the share of AI workers (based on the resume data) from 2012 to 2017, standardized to mean zero and standard deviation of one. Regressions are weighted by the total (industry-level) number of Cognism resumes. The dependent variables are the changes, from 2012 to 2017, in log industry sales in columns 1 and 2, log industry employment in columns 3 and 4, the Herfindahl-Hirschman Index (HHI) in columns 5 and 6, and the market share of the top 4 firms in an industry in columns 7 and 8. All specifications control for industry sector fixed effects. Regressions in columns 2, 4, 6, and 8 also include industry-level controls for log total employment, log total sales, and log average wage in 2012. Standard errors are robust against heteroskedasticity and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Log Sales		Log Employment		HHI		4 Largest Firms Sales Share	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ Share AI Workers	0.053* (0.030)	0.084** (0.040)	0.038 (0.029)	0.056* (0.031)	0.046*** (0.014)	0.049*** (0.016)	0.013** (0.005)	0.017*** (0.005)
Industry Sector FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	N	Y	N	Y	N	Y	N	Y
Observations	246	246	246	246	88	88	237	237